# CREDIT RISK ANALYSIS SYSTEM

A data-driven approach to understanding loan default risk

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Data Science Capstone Project

# 1. Project Overview

What the system does and why it matters

### Problem

Loan default can create financial losses.
Reviewing applicant data manually can take time.

### Objective

Analyze loan data, explore default patterns, compare ML models, and estimate risk for a new applicant.

### Project outcome

An interactive dashboard that combines dataset summaries, graphs, model comparison, and applicant risk prediction.

# 2. Dataset & Preprocessing

The project uses a 5,000-row credit risk dataset

### Dataset size

5,000 applicants
12 original columns

### Target column

loan_status
0 = No default
1 = Default

### Feature examples

Income, age, loan amount, interest rate, employment, loan purpose, credit history

## Preprocessing steps

- Handle missing values using suitable imputation
- Remove duplicate rows
- Scale numeric features
- Encode categorical features for machine learning

# 3. Exploratory Data Analysis

Dashboard graphs help reveal patterns in the applicant data

### Loan status distribution

3,611 applicants had no recorded default; 1,389 had a default.

### Income distribution

Income values are concentrated at lower-to-middle ranges, with a right-skewed tail.

### Loan amount distribution

The graph shows how loan amounts vary across the dataset.

### Home ownership

The dashboard compares applicant counts across home-ownership categories.

**Overall default rate: 27.78% of the 5,000 records**

# 4. SQL-Based Analysis

SQL queries summarize risk patterns by group

### Overall risk summary

Total applicants: 5,000
Defaults: 1,389
Non-defaults: 3,611
Default rate: 27.78%

### Loan purpose

Compare applicant totals and default percentages for education, venture, medical, personal, debt consolidation, and home improvement.

### Home ownership

Group applicants by RENT, MORTGAGE, OWN, and OTHER to compare default counts and percentages.

Database file: credit_risk.db | Summary outputs: risk_summary.csv, purpose_summary.csv, ownership_summary.csv

# 5. Machine Learning Models

Three classification algorithms were trained and evaluated

### Logistic Regression

A baseline classification model that estimates the relationship between features and default.

### Decision Tree

Uses feature-based decision rules. It achieved the highest F1-score in this comparison.

### Random Forest

Combines multiple decision trees to make a prediction.

Evaluation metrics: Accuracy • Precision • Recall • F1-score • ROC-AUC
Preprocessing: median imputation and scaling for numeric features; most-frequent imputation and one-hot encoding for categorical features.

# 6. Model Performance Comparison

Test-set metrics from the current project run

| Model | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.726 | 0.526 | 0.144 | 0.226 | 0.640 |
| **Decision Tree** | 0.604 | 0.337 | 0.439 | 0.381 | 0.542 |
| Random Forest | 0.704 | 0.412 | 0.151 | 0.221 | 0.596 |

**Key finding**

Decision Tree has the highest F1-score (0.381). Logistic Regression has the highest accuracy (0.726) and ROC-AUC (0.64

# 7. Interactive Dashboard

Built with Gradio and organized into three tabs

### Dataset Analysis

Displays applicant totals and visualizations for loan status, income, loan amount, and home ownership.

### Model Comparison

Shows accuracy, precision, recall, F1-score, and ROC-AUC for all three models.

### New Applicant Prediction

Accepts applicant details and displays predicted risk, estimated default probability, and a recommendation.

Add your actual dashboard screenshots to this slide if your presentation template allows; keep text readable.

# 8. Applicant Risk Prediction

Example output from the running dashboard

**Example input**

Age: 25
Annual income: 40,000
Home ownership: RENT
Employment length: 2 years
Loan purpose: EDUCATION
Loan amount: 5,000
Interest rate: 12.5%
Previous default: No
Credit history: 3 years

**Example prediction**

Predicted class: No default
Estimated default probability: 32.30%
Risk label: Lower predicted default risk
Recommendation: Further review before any loan decision

**This is a model estimate, not a guarantee. Final lending decisions require human review and additional checks.**

# 9. Conclusion & Future Improvements

### Conclusion

- Analyzed 5,000 loan records
- Used graphs and SQL summaries
- Compared three ML models
- Built an interactive dashboard
- Added new-applicant risk prediction

### Future improvements

- Improve recall and overall model quality
- Test on larger and more varied datasets
- Add more model validation
- Deploy with a stable public URL
- Improve explanations for risk predictions

# Thank you

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